

SECV3263 MULTIMEDIA WEB PROGRAMMING

SEMESTER 2 2025/2026

ALTERNATIVE ASSESSMENT – INDIVIDUAL SELF REFLECTION

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Role: About Us Section Developer and CSS Stylist

I developed the “About Us Section” page of our group’s interactive 3D Portfolio Website. This involve creating profile cards for all four members that display each member’s name, role, education, career interests, and personal achievements. I also built floating 3D shapes arranged in a circular formation, with connecting lines. I implemented hover interaction where shapes scale up and emit a glow, as well as ensuring the fog and floor colors update dynamically when switching between light and dark themes. I also assisted with CSS design tokens and responsive styling to ensure the section adapts well to different screen sizes.

The main challenge I encountered was making the section fully theme-aware. Initially, the fog and floor colors were hardcoded, so they looked out of place when toggling between the light and dark themes. I solved this by detecting the active theme using `document.body.classList.contains('dark-theme')` and adding a `themeChanged` event listener to update the colors in real-time. Positioning the profile cards without overlapping the 3D shapes while remaining readable on mobile devices was also tricky. Eventually, I tweaked the CSS transform to center the cards and added media queries to adjust card sizes and spacing on smaller screens.

Through this project, I gained practical experience in Three.js geometry creation, responsive 3D design, theme-aware development using CSS custom properties with JavaScript listeners, and raycaster-based hover interactions. I also learnt using Git branching and pull requests for collaborative development. This mirrors professional software engineering development workflows. I would improve this assessment further by adding click-to-expand profile details, optimizing performance for lower-end mobile devices that improves accessibility with ARIA labels as well as implementing smooth camera transitions between sections. Overall, this alternative assessment deepened my understanding of Three.js and responsive design. I look forward to applying these skills in future work.